

SMART ENERGY CONSUMPTION PREDICTION

Forecasting daily household electricity usage and flagging high-usage days

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XGBoost regression • UCI Household Power Consumption + Weather Data

Problem Statement

Day-to-day electricity demand is hard to predict — it depends on routine behaviour, calendar effects, and weather that interact non-linearly. Knowing in advance when consumption will spike lets providers balance supply and demand, and lets households shift discretionary, high-load activities to off-peak periods.

The goal: build a forecast accurate enough to be useful — and understand, rather than simply report, where and why it fails.

Energy Providers

Balance supply and demand more efficiently ahead of spikes.

Households

Shift laundry, dishwashing, and water heating to off-peak periods.

Dataset & Approach

DATASET

Power: UCI Individual Household Electric Power Consumption — minute-level readings, single household near Paris, Dec 2006–Nov 2010.

Weather: historical daily temperature for the same location/period, from Open-Meteo.

Also: SHAP for feature attribution, error-vs-sub-metering investigation, and a threshold-based high-usage alert layer.

1

Clean & resample

Minute-level readings resampled to daily totals.

2

Engineer features

Calendar, lag, rolling mean/std, and a “heating degree” weather feature.

3

Train & tune

XGBoost regression, tuned via RandomizedSearchCV with time-aware cross-validation.

4

Benchmark honestly

Compared against “same as yesterday” / “same as last week” naive baselines.

Results

Tuned XGBoost vs. naive baselines on a held-out time period

Model	MAE	RMSE
Naive — same as yesterday	286.29	405.35
Naive — same as last week	387.99	532.19
Tuned XGBoost (this project)	247.80	338.54

~13–17% improvement over the best naive baseline on both MAE and RMSE
— modest but real, since recent consumption is already a strong predictor.

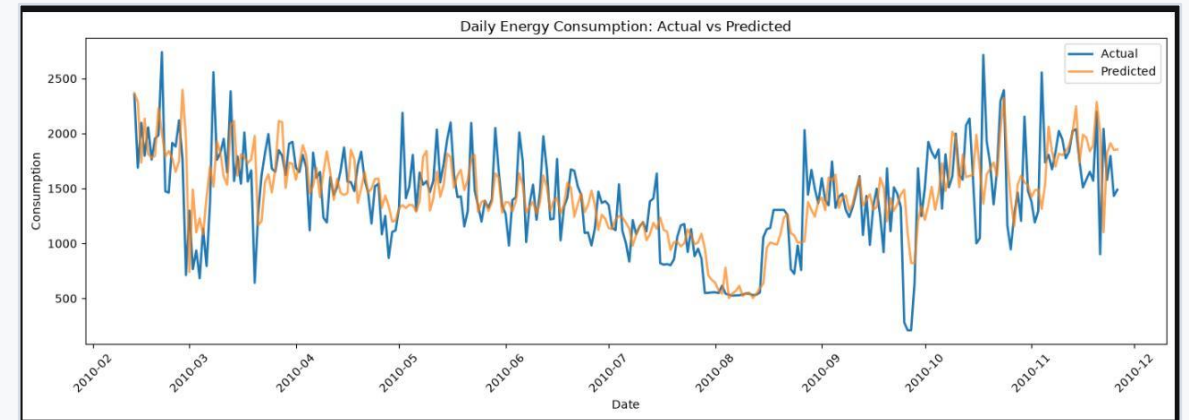


Figure 1. Actual vs. predicted daily energy consumption over the evaluation period.

Recent usage dominates

Yesterday's value and the 7-day rolling average drive most predictions.

Weekends run hotter

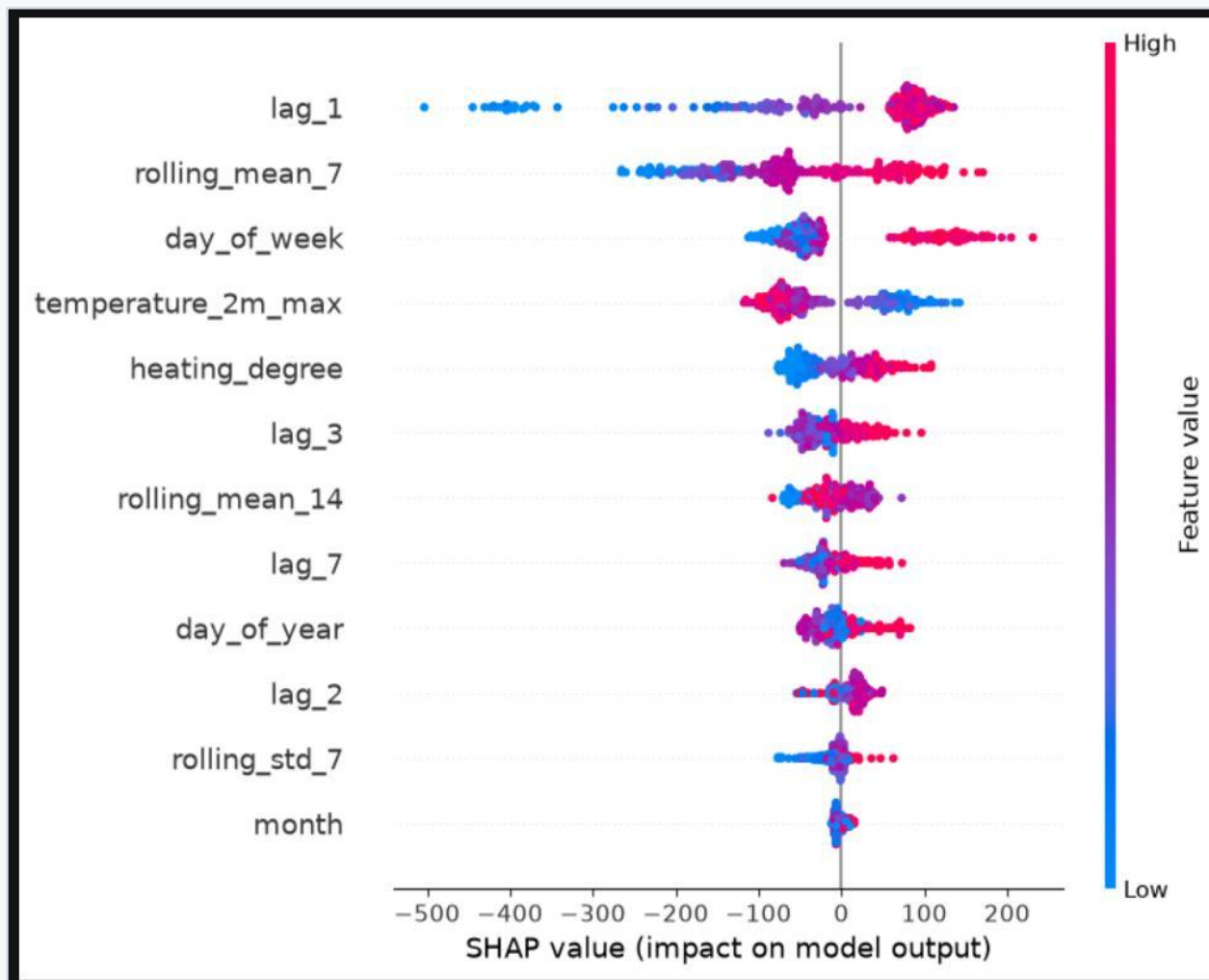
Consumption is ~17% higher on Saturdays & Sundays than weekdays.

Weather effect, engineered

A "heating degree" feature beats raw temperature and behaves physically.

Feature Importance (SHAP)

Moving beyond a ranking to understand direction and consistency of each feature's effect



- lag_1 (yesterday) and rolling_mean_7 are the two strongest drivers.
- High recent consumption (red) consistently pushes predictions upward; low recent consumption pushes them down.
- The relationship is monotonic — a useful sanity check that the model behaves as expected.
- day_of_week and temperature-related features follow as secondary drivers.

Figure 2. SHAP summary plot showing each feature's impact on model output, ranked by importance.

Error Analysis

Residuals were examined over time; the ten largest errors were cross-checked against sub-metering data

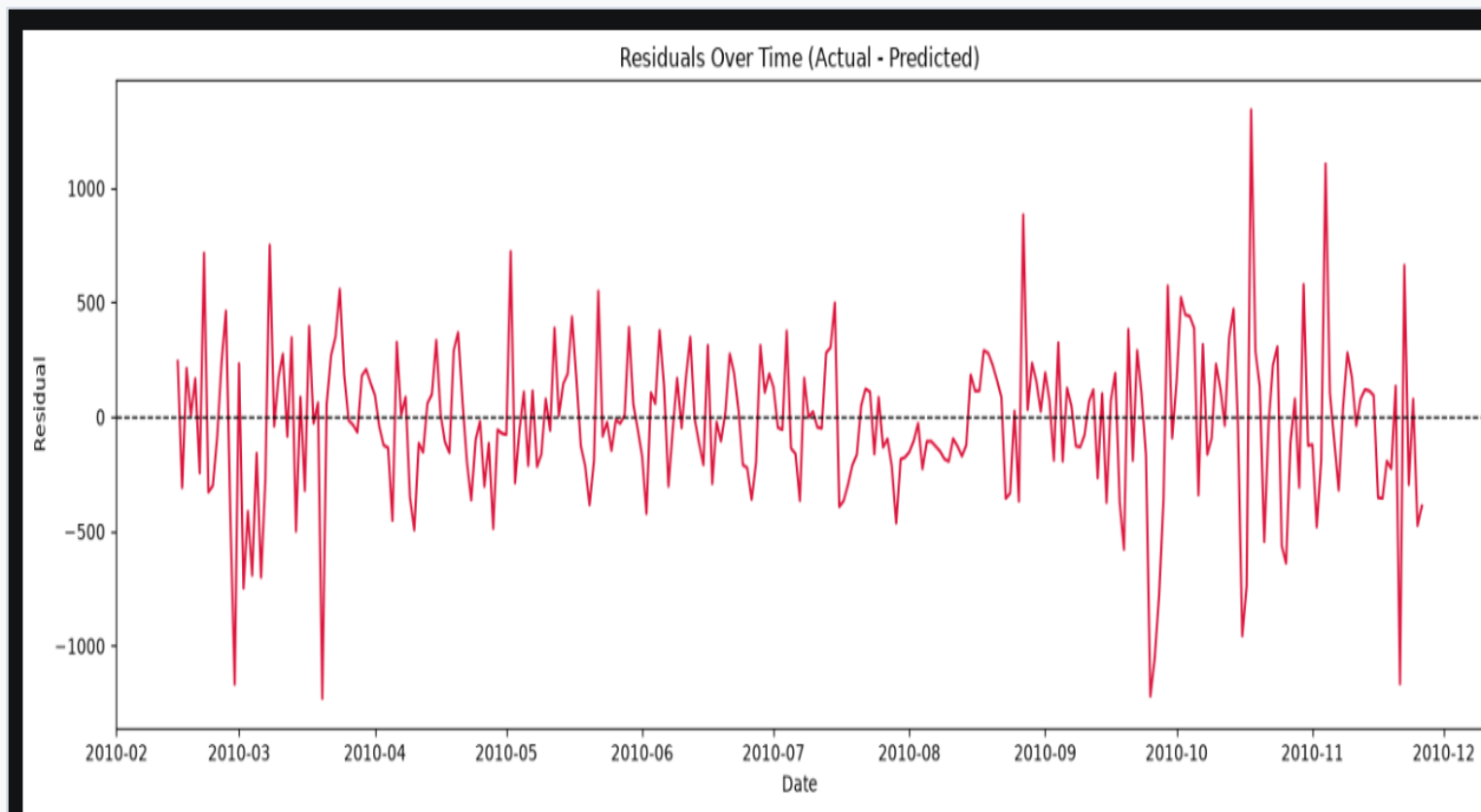


Figure 3. Residuals (actual minus predicted) over the evaluation period.

8 / 10

largest errors traced to occupancy anomalies (travel or guests)

Residuals centre near zero with no long-term drift — the model isn't systematically biased over time, though spikes align with specific high-error days.

Not observable in the data: none of these occupancy events could be anticipated from any available feature — a genuine data limitation, not a model weakness.

High-Usage Day Alerts

True high-usage days are rare (~2%) — threshold choice is a precision/recall trade-off, not a single “correct” number

67%

RECALL

of true high-usage days are caught by the alert

~9%

PRECISION

of flagged days are true high-usage days

Threshold: 85th percentile of predicted consumption, chosen after testing multiple thresholds explicitly. For a low-stakes advisory tool, missing a real spike costs more than an occasional false alarm — so recall was favoured by design.

Limitations



Single household

Trained on one household — won't generalise elsewhere without retraining.



No occupancy signal

Whether people are home, travelling, or hosting is unobserved — the single largest source of error.



Limited history

Only ~4 years of data, so not every seasonal extreme has been seen.



Recall-favoured alerts

The alert system is tuned for recall over precision — worth reconsidering for a more sensitive use case.

Conclusion

A fairly standard gradient-boosted model, combined with careful feature engineering and honest evaluation against baselines, meaningfully improves on naive forecasts of household electricity consumption.

Just as importantly, digging into where the model is wrong — rather than stopping at a single accuracy number — surfaced a concrete, well-evidenced limitation (unobserved occupancy) that no amount of tuning could fix, and clarified the real trade-offs in turning a forecast into an actionable alert.

247.80

MAE (vs. 286.29 baseline)

13–17%

improvement over naive

67%

high-usage days caught